

Year 12 Mathematics Term II Examination 2007

QUESTION 1.

Marks

- | | | |
|-------|---|----------|
| (a) | Solve for x : $-2x^2 + 1 \geq -31$ | 2 |
| (b) | If α and β are the roots of the equation $4x^2 - 3x + 5 = 0$ find : | |
| (i) | $\alpha + \beta$ | 1 |
| (ii) | $\frac{1}{\alpha} + \frac{1}{\beta}$ | 2 |
| (iii) | $\alpha^2 + \beta^2$ | 2 |
| (c) | (i) State the condition for equal roots of a quadratic equation. | 1 |
| | (ii) The line with equation $y = mx - 9$ is a tangent to the parabola with equation $y = x^2$. Find the values of m . | 2 |
| (d) | The acceleration $\ddot{x}$ of a particle is given by $\ddot{x} = 15\sqrt{t}$, where the displacement is x metres and time t seconds. If the particle is initially 4 metres to the left of the origin and has an initial velocity of -3 metres per second, find the velocity and displacement functions in terms of time t . | 2 |
| (e) | Two dice each with the numbers 2, 4, 6, 8, 10 and 12 on their faces are thrown. By using a dot diagram or otherwise, find the probability that the sum is greater than 10. | 3 |

QUESTION 2.

- | | | |
|-----|--|----------|
| (a) | Write a quadratic equation with the sum of roots equal to 6 and the product of roots equal to -4 . | 2 |
|-----|--|----------|

A particle moves horizontally in a straight line. The displacement x metres as a function of time t seconds is given by $x = 4t^3 - 15t^2 - 18t - 6$

- | | | |
|-----|--|----------|
| (b) | (i) Find the initial displacement, initial velocity and initial acceleration. | 3 |
| | (ii) Find the time and displacement when the particle comes to rest. | 3 |
| | (iii) Determine the direction of motion after the particle comes to rest. Justify your answer. | 2 |

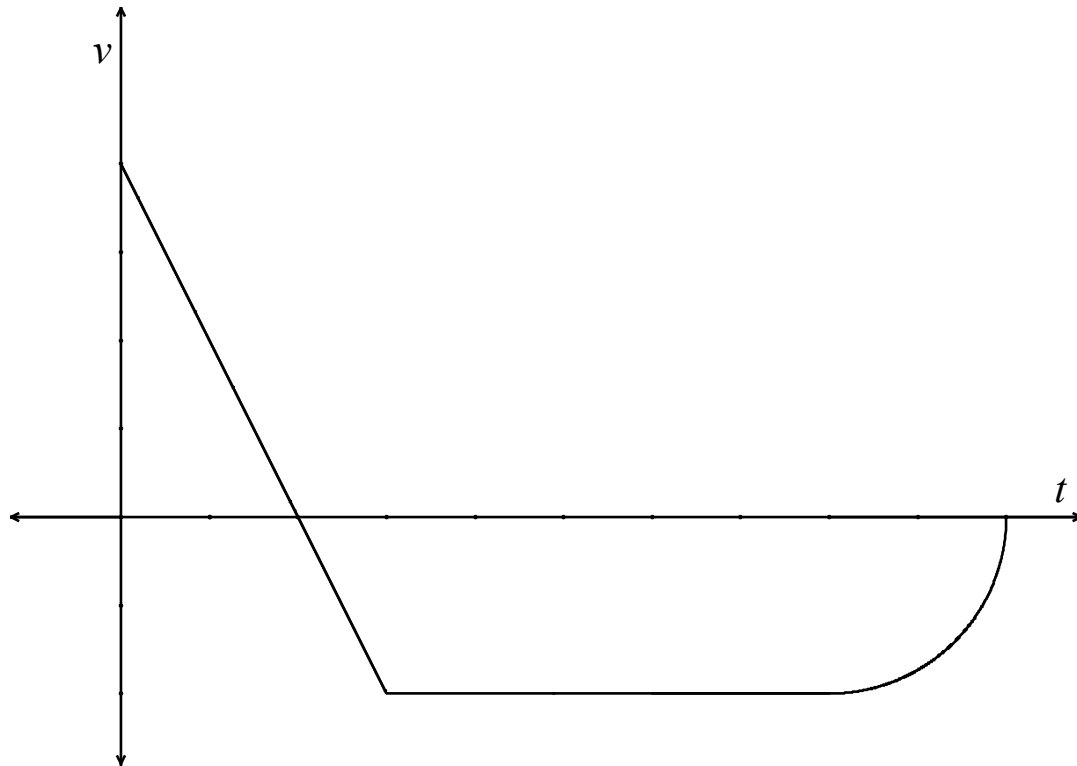
As a sliding door closes in time t seconds, the door opening x cm is given by $x = Ae^{-kt}$.

- | | | |
|-----|---|----------|
| (c) | (i) Show that the rate of closure of the door $\frac{dx}{dt}$ is proportional to the size of the door opening x . | 1 |
| | (ii) If the initial opening is 80 cm and the initial speed of closure is 10 cm/s, find the values of A and k . | 2 |
| | (iii) Find the time(correct to 2 decimal places) for the door to be 80% closed. | 2 |

QUESTION 3.**Marks**

- (a) The velocity v m/s against time t seconds graph shown is comprised of straight lines and a circular arc.

- (i) Copy the graph, hence graph directly beneath using the same time scale :

1

- (ii) (α) the acceleration function $\ddot{x}$ against time t for $0 \leq t \leq 10$.

2

- (β) the displacement function x against time t for $0 \leq t \leq 10$ if the particle is initially at the origin. (Use scale $1 \text{ cm} = 2 \text{ m}$)

3

- (b) A car travels at 100 km/hr along a straight horizontal road at night. Posts are placed at 5 m intervals along the left side of the road.

- (i) Find the speed of the car in metres per second (correct to 2 decimal places).

1

- (ii) A truck with bright lights approaches the car from the opposite direction and blinds the driver of the car for 2 seconds.

- (α) Find the distance the car travels while the driver is blinded, hence show that the number posts N that the car could pass is 12.

2

If the probability of not hitting a post is $\frac{99}{100}$, find the probability of :

- (β) not hitting the first three posts (4 decimal places).

2

- (γ) hitting at least one post while the driver is blinded (4 decimal places).

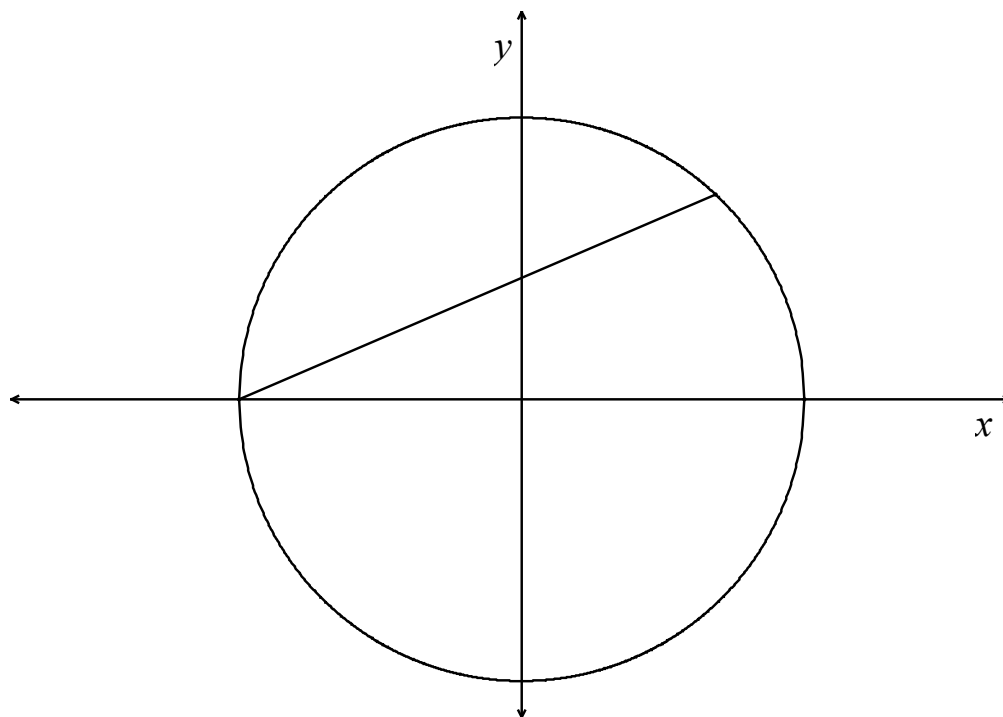
2

QUESTION 4.**5**

- (a) The displacement function of a particle is given by : $x = \sqrt{1+t^2}$.

By finding $\ddot{x}$ show that the direction of the force applied to the particle never changes.

- (b)



The chord shown cuts the circle with equation $x^2 + y^2 = 1$ at the points A and P .

- | | | |
|-------|---|----------|
| (i) | Find the equation in gradient intercept form, of the chord AP with gradient m . | 1 |
| (ii) | Show that the x values of the intersection points A and P satisfy the equation $(m^2 + 1)x^2 + 2m^2x + m^2 - 1 = 0$ | 2 |
| (iii) | Find the co-ordinates of the intersection point P . | 3 |
| (iv) | If $\angle PAO = \alpha$ show $\angle POB = 2\alpha$, giving reasons. | 2 |
| (v) | Hence find an expression for $\tan 2\alpha$ in terms of $\tan \alpha$. | 2 |

End Of Examination

SOLUTIONS

(a) $-2x^2 + 1 \geq -3$
 $-2x^2 \geq -32$
 $x^2 \leq 16$
 $-4 \leq x \leq 4$ ①

b) (i) $2 + \beta = \frac{3}{4}$ ①

(ii) $\frac{1}{2} + \frac{1}{\beta} = \frac{2 + \beta}{2\beta}$
 $= \frac{3}{4}$
 $\frac{5}{4}$ ①
 $= \frac{3}{5}$ ①

(iii) $2^2 + \beta^2 = (2 + \beta)^2 - 2 \times 2 \times \beta$ ①
 $= \left(\frac{3}{4}\right)^2 - 2 \times \frac{5}{4}$
 $= -\frac{31}{16}$
 $= -1\frac{15}{16}$ ①

c) (i) $\Delta = 0$
(ii) $x^2 - mx + 9 = 0$
 $\Delta = 0$
 $m^2 - 36 = 0$ ①
 $m = \pm 6$ ①

d) $x'' = 15\sqrt{x}$
 $\dot{x} = 10x^{\frac{3}{2}} + C$
 $\dot{x} = -3$ at $x = 0$
 $-3 = 0 + C$
 $C = -3$
 $\dot{x} = 10x^{\frac{3}{2}} - 3$ ①
 $x = 4x^{\frac{5}{2}} - 3x + C$
 $x = -4$ at $x = 0 \Rightarrow C = -4$
 $\therefore x = 4x^{\frac{5}{2}} - 3x - 4$ ①

	2	4	6	8	10	12
2						
4						
6						
8						
10						
12						

$P(\text{sum} > 10) = \frac{26}{36} = \frac{13}{18}$ ①

2(a) $x^2 - 6x - 4 = 0$ ①

b) $x = 1x^3 - 15x^2 - 18x - 6$
 $v = 12x^2 - 30x - 18$
 $\dot{x} = 24x - 30$

(i) $t = 0$ $x = -6$ m [6 m left of 0] ①
 $v = -18$ m/s [18 m/s left] ①
 $\dot{x} = -30$ m/s² ①

(ii) $v = 0$
 $12x^2 - 30x - 18 = 0$
 $2x^2 - 5x - 3 = 0$
 $(2x + 1)(x - 3) = 0$ ①
 $x = -\frac{1}{2}$ or $x = 3$
 But $x > 0 \therefore x = 3$ s only ①
 $x = -87$ m. ①

(iii) $x'' = 24 \times 3 - 30$
 $= 42$
 > 0 ①
 i. Force $mx'' > 0$ as $m > 0$ & $x'' > 0$
 Particle moves right from rest, never stops or changes direction after $t > 3$

$$20(i) \quad x = Ae^{-kt}$$

$$\frac{dx}{dt} = -kAe^{-kt}$$

$$\frac{dx}{dt} = -kx$$

$$\therefore \frac{dx}{dt} \propto (-x)$$

$$t=0 \quad x=80 \Rightarrow 80 = Ae^0$$

$$A=80 \quad (1)$$

$$\text{Also from } -10 = -k \cdot 80$$

$$k = \frac{1}{8}$$

$$\therefore x = 80e^{-\frac{t}{8}}$$

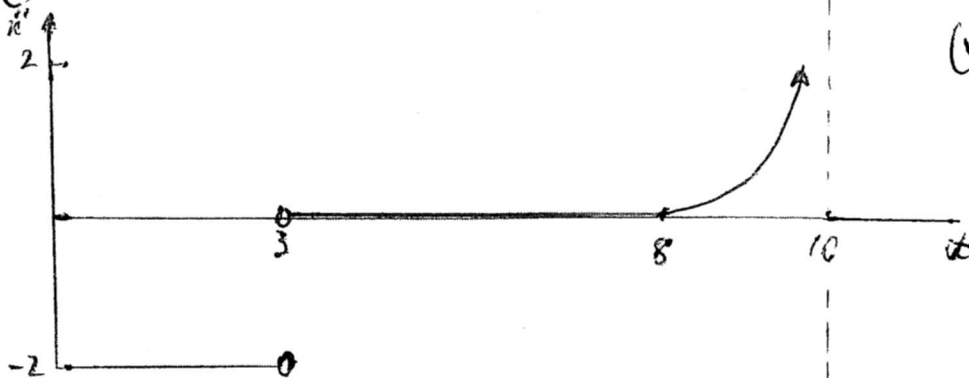
$$(ii) \quad 0.24 = Ae^{-\frac{t}{8}}$$

$$\frac{t}{8} = -\ln 0.2$$

$$t = 12.88 \text{ s}$$

3(a) (i) (1)

(ii)



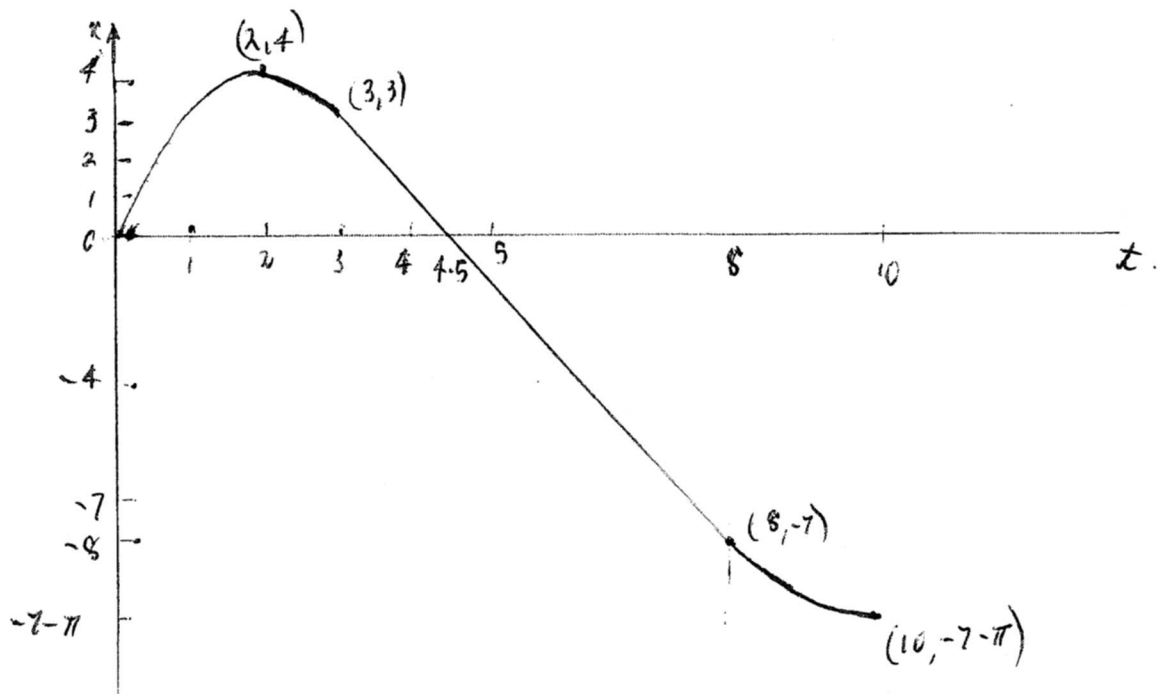
(iv) Time return to origin 4.5 s

(v) Average speed

$$= \frac{2+4+7+\pi}{10}$$

$$= 1.8 \text{ m/s}$$

(iii)



$$(4) (i) \quad v = \frac{100 \times 1000}{3600}$$

$$= 27.78 \text{ m/s} \quad (1)$$

$$(ii) (2) \quad \text{Distance travelled} = 2 \times 27.78$$

$$= 55.56 \text{ m}$$

$$\text{Number posts} = \frac{55.56}{5} + 1$$

$$(BS) \quad P(\tilde{H} \tilde{H} \tilde{H}) = 0.99^3 \quad (2)$$

$$= 0.9703 \quad (1)$$

$$(BS) \quad P(\text{Hitting}) = 1 - P(\text{Not Hitting}) \quad (1)$$

$$= 1 - 0.99^3$$

$$= 0.1136 \quad (1)$$

$$4(a) \quad x = \sqrt{1+t^2}$$

$$\dot{x} = \frac{t}{\sqrt{1+t^2}} \quad (2)$$

$$\ddot{x} = \frac{\sqrt{1+t^2} \cdot 1 - t \cdot \frac{t}{\sqrt{1+t^2}}}{1+t^2}$$

$$= \frac{1+t^2 - t^2}{(1+t^2)^{3/2}}$$

$$= \frac{1}{(1+t^2)^{3/2}}$$

$$> 0 \quad \text{for all } t > 0 \quad (2)$$

$$\text{Since } F = m\ddot{x} \quad m > 0$$

$\therefore$ direction force always > 0
and never equals zero

$\therefore$ Force never changes direction (1)

(b) (i) $\angle PAO = \alpha$ given

$\angle OPA = \alpha$ [equal angles opposite equal sides/radii in ΔAPO] ①

$\angle POB = \angle PAO + \angle OPA$ [Exterior angle triangle is equal to the sum of the two interior opposite angles] ①

$= \alpha + \alpha$

$= 2\alpha$

(ii) $y - y_1 = m(x - x_1)$

$y - 0 = m(x + 1)$

$y = mx + m$ ①

(iii) $x^2 + (mx + m)^2 = 1$ ①

$x^2 + m^2x^2 + 2m^2x + m^2 = 1$ ①

$(1+m^2)x^2 + 2m^2x + m^2 - 1 = 0$

$(x+1)((1+m^2)x + m^2 - 1) = 0$ ①

$x = -1$ or $x = \frac{1-m^2}{1+m^2}$ ①

But $x \neq -1$ $\therefore x_P = \frac{1-m^2}{1+m^2}$ $y_P = m\left(\frac{1-m^2}{1+m^2}\right) + m$

$\therefore y_P = \frac{m - m^3 + m^3 + m}{1+m^2}$

$= \frac{2m}{1+m^2}$ ①

But $m = \tan \alpha$

$\therefore \tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$ ①

$P \equiv \left(\frac{1-m^2}{1+m^2}, \frac{2m}{1+m^2} \right)$